## 10.1 Secondary Growth in Dicot Stem

### Vascular Cambium

The vascular cambium is the lateral meristem that produces the secondary vascular tissues. i.e., secondary xylem and secondary phloem.

### Origin and Formation of Vascular Cambium

A strip of vascular cambium that is believed to originate from the procambium is present between xylem and phloem of the vascular bundle. This cambial strip is known as **intrafascicular or fascicular cambium**. In between the vascular bundles, a few parenchymatous cells of the medullary rays that are in line with the fascicular cambium become meristematic and form strips of vascular cambium. It is called **interfascicular cambium**.

This interfascicular cambium joins with the intrafascicular cambium on both sides to form a continuous ring. It is called a **vascular cambial ring.** The differences between interfascicular and intrafascicular cambia are summarised below:

| Intrafascicular cambium | Interfascicular cambium |
|---|---|
| Present inside the vascular bundles | Present in between the vascular bundles. |
| Originates from the procambium. | Originates from the medullary rays. |
| Initially it forms a part of the primary meristem. | From the beginning it forms a part of the secondary meristem. |

### Organization of Vascular Cambium

The cells of vascular cambium do not fit into the usual description of meristems which have isodiametric cells, with a dense cytoplasm and large nuclei. While the active vascular cambium possesses cells with large central vacuole (or vacuoles) surrounded by a thin, layers of dense cytoplasm.

Further, the most important character of the vascular cambium is the presence of two kinds of initials, namely, **fusiform initials** and **ray initials**.

#### Fusiform Initials

These are vertically elongated cells. They give rise to the longitudinal or axial system of the secondary xylem (treachery elements, fibres, and axial parenchyma) and phloem (sieve elements, fibers, and axial parenchyma).

Based on the arrangement of the fusiform initials, two types of vascular cambium are recognized.

### Storied (Stratified cambium) and Non-Storied (Non-stratified cambium)

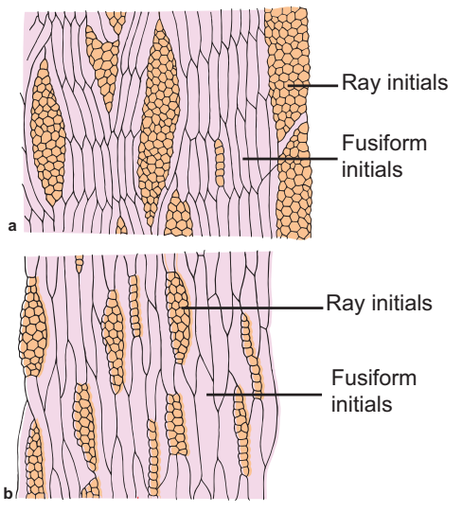



**Figure 10.2:** Tangential longitudinal section (TLS) of cambium (a) Storied cambium (b) Non-storied cambium

If the fusiform initials are arranged in horizontal tiers, with the end of the cells of one tier appearing at approximately the same level, as seen in tangential longitudinal section (TLS), it is called **storied (stratified) cambium**. It is the characteristic of the plants with short fusiform initials. Whereas in plants with long fusiform initials, they strongly overlap at the ends, and this type of cambium is called **non-storied (non-startified) cambium**.

#### Ray Initials

These are horizontally elongated cells. They give rise to the ray cells and form the elements of the radial system of secondary xylem and phloem.



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